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IOT Based Home Automation

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Abstract : We are living in a new era of Computer Technology which is known as Internet Of Things (IOT). IOT plays an important role as it consists of smart machines interacting and communicating with other machines, objects, environments and infrastructures. Each device is provided with unique identifiers and has the ability to transfer data over a network without human - to - human or human - to - computer interaction. With the advancement in technologies IOT and smart home have come together to emerge as a smart technology. In this era, we can't imagine any smart home without IOT. Using IOT, it is possible to make all the electronic gadgets smarter by connecting them through the Internet. System is made more user friendly by using android and web technologies. Home automation can access and control all the home equipment from each and every corner of the world. It is basically based on a multimodal application which can be operated using voice recognition, the user can operate it by giving commands through Google Assistant or any web based Application. This device can also be very beneficial for physically handicapped people to control home appliances using mobile applications. The main objective is to make our system more secure and intelligent.

IndexTerms - Home Automation System, Relay, Node MCU (ESP8266), IFTTT, Adafruit, Internet of Things (IoT) Wireless Sensor Network, MQTT.

I. INTRODUCTION

Due to advancement in technologies, Human - Machine Interaction has become more realistic in day - to - day life [1]-[3]. Earlier the Internet was just used for communication purposes but today, after the HMI research it can be used for things as well [4]-[6]. Its purpose is to connect anything to the Internet and can access it from any part of the world.

IOT has shown its contribution in all types of applications from small scale applications to large scale applications. IOT has provided all those applications which can convert non - smart devices into smart devices [13]-[16].

IOT is basically a concept in which every device is assigned an IP Address and with the help of that IP Address that device is identifiable on the Internet. After the advancement in technologies, wireless controlling environments like Bluetooth, Wi-Fi have allowed different devices to connect with each other. Wi-Fi shields work as a standalone device which acts as a micro web server which has eliminated the wired connection between the arduino board and computer. Wi-Fi shields connect to the Internet using a wireless router or wireless hotspot which acts as a gateway for Aurdino and the Internet to communicate with each other.

II. HOME AUTOMATION SYSTEM

Home automation systems help us to use our devices or appliances automatically by automatic sensation and have an electronic control over household features, activities and appliances [21]-[24]. By replacing all the switch boards present in different rooms, operating each device is made convenient for the members of the home, especially for the elders and the physically handicapped which uses voice as well as web based services for controlling the home appliances [9], [25]. Also, we can add security by installing cameras in the home, to trace all the activities. Also, we can save electricity and electricity bills by monitoring their home and turning ON/OFF the appliances.

By making the system more intelligent, we can prevent the intruder from entering the house using sensors and cameras and also we can detect the presence of the person. As soon as the person is detected, lights and fan will turn ON automatically.

Wireless Sensor Networks (WSN) is a network device incorporated by the Home Automation System for gathering information from various sensors wirelessly over the network. WSN establishes various sensors by which we can have a control over the occurrence of specific devices and their activities. To reduce human efforts, WSN based HAS is developed especially for elderly

people and physically handicapped by providing them an intelligent control over an app-based application at an affordable price with just a single touch we can monitor and control fans and lights of our home, it is also called SMART TOUCH. It ensures us to provide comfortability and convenience.

Home Automation uses MQTT for sending/ receiving data from the sensor. For accessing the data from the sensors Raspberry Pie acts as a gateway and with the help of web- based interface users can control all the home appliances. The system is designed using ZigBee, as it offers low- power operation, high security, and robustness to complex systems.

III. SYSTEM DESIGN AND IMPLEMENTATION

The inputs from the voice mode are the most important inputs for human - machine interaction. We can make our house more user friendly by using Google Assistant along with web based applications to control all the home appliances [32].

The model can be handled in a multimodal manner, that means we can handle it manually as well as automatically. The advantage is that in the noisy background the quality of web - based Google Assistant degrades at that time the appliances can be handled manually. It enhances the robustness of the device which makes it more flexible.

We can convert the non - intelligent devices into intelligent devices, i.e we can implement a smart home with a main controller unit which is connected with 24-hour available wi-fi. We can also ensure that wi-fi connections do not turn off, for that main controller is programmed to establish automatic wi-fi connection with the available network and also to make it connect to an automated power backup.

All the subunits are connected with the main unit so that the devices which are not smart can be made intelligent. So that the smart home can be controlled and accessed by their users using the Blynk IOT Application which is a web-based service.
System Requirement

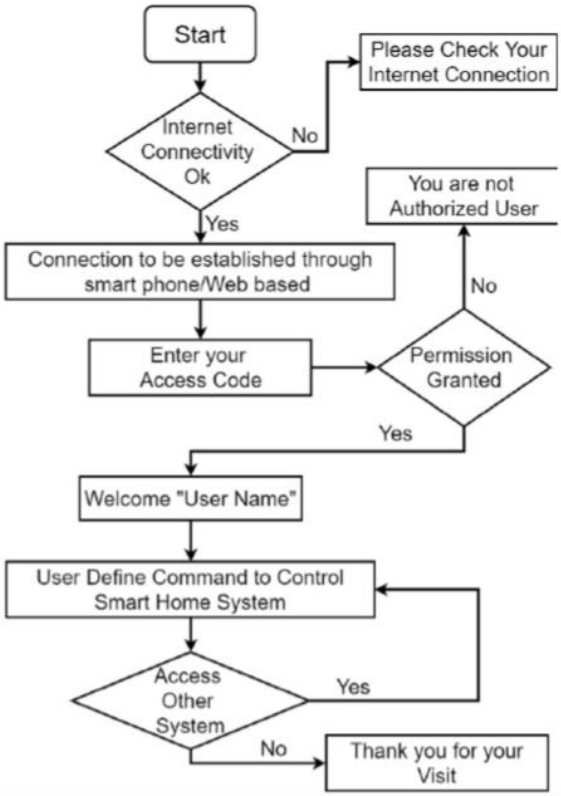
- Node MCU (ESP8266)
- IFTTT
- Adafruit
- Arduino Software IDE

Node MCU has been selected as a controller of home automation systems as it is an open source firmware which provides the flexibility to develop IOT based applications [33]. This firmware has a basic programming environment and a quick and easy scripting language, due to its low cost and wi-fi enabled feature, it has gained a lot of popularity. It provides NodeJS to perform tasks as it requires less computation time and uses Lua Script, thus making the system operate much faster.

IFTTT (If This Then That) is an interface which provides web-based services to the devices which are connected to the mobile application. It makes it easier for the devices to work on the basis of mobile applications using conditional statements [34].

Adafruit acts as an MQTT broker as it is a library which supports Message Queue Telemetry Telegram (MQTT) [28], [35]. Basically, MQTT is a protocol service that helps in sending and receiving the feed data. The advantage of using MQTT is that it provides a faster rate for transmission of the data and makes connectivity with fewer data bytes. It requires 80 bytes to connect the device to the server and 20 bytes from server to device. We are using Arduino Software IDE to compile the code [36].

IV. WORKING MODEL



Smart Home Automation System Flow

The flow chart shows the working of the Home Automation System. The first step is to initialize the wi-fi. This can be done by writing the network name and password in the code and that code will be uploaded to Node MCU. One can access their smart home either using Google assistant or any web-based service.

For controlling/monitoring smart home Google assistant is used and if there is a noisy background then we can use a web based service. Google assistant will always ask for the security code provided to it, which will prevent unauthorized access.

CONCLUSION AND FUTURE WORK

Thus, with the help of the device control unit, it has become easier and possible to convert home appliances into smart and intelligent devices using IOT. The proposed Home Automation System is low cost, practical and simple to use. By implementing such a system, we can easily monitor and access the smart home from anywhere, and also we can ensure conservation of energy. The whole system is designed as a helping hand for the old aged and differently abled person.

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